

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Description about the assessment tool:

Short-answer questions effectively measure conceptual understanding, logical reasoning, and the ability to apply theoretical knowledge without requiring lengthy descriptive responses. This assessment method is also efficient for evaluating multiple OSI-related concepts within a limited time while ensuring objective and consistent evaluation.

Code of Conduct:

I S. Nithresh (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

S. Nithresh
Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	2	2	2
Explanation	25%	2	2	2	2
Application	20%	2	2	2	2
Organization	15%	1	1	1	1
Accuracy	10%	1	1	1	1
Clarity	5%	2	2	2	2

Total Marks 35 /40

[Signature]
Signature of the Faculty

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted,

and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and	Mostly accurate with minor	Partially correct with	Incorrect answers with

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		covers all required points	omissions	missing points	major gaps
Clarity	5%	Clear, neat, professional presentation	Understandabl e with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

⑥ Presentation Layer - Compression & Encryption:

Given:

* Original file size = 900 MB

* Compression = 40%

* Encryption overhead = 5% of compressed size.

Step 1: Calculate compressed file size

compression reduces the file by 40%.

$$\text{Compressed size} = 900 \times (1 - 0.40) = 900 \times 0.60 = 540 \text{ MB}$$

Step 2: Calculate encryption overhead

Encryption overhead = 5% of 540

$$= \frac{5}{100} \times 540 = 27 \text{ MB}$$

Step 3: Calculate encrypted file size

$$\text{Encrypted file size} = 540 + 27$$

$$= 567 \text{ MB}$$

Step 4: Calculate total percentage reduction

$$\text{Reduction} = 900 - 567 = 333 \text{ MB}$$

$$\text{Percentage reduction} = \frac{333}{900} \times 100 = 37\%$$

The permutation Layer compresses the original 900 MB file to 567 MB. After encryption, the final file size becomes 567 MB. Therefore, the total reduction compared with the original file is 37%.

Application Layer - FTP Data Transfers:

Given:

* Total data = 3GB

* Throughput = 120 Mbps

* Each request = 3MB

Step 1: Calculate total numbers of requests

$$3\text{GB} = 3 \times 1024$$

$$3\text{G} = 3072 \text{ MB}$$

$$\text{Number of requests} = \frac{3072}{3}$$

1024 requests

Step 2: Convert data into megabits

$$3\text{GB} = 3 \times 1024 \times 1024 \times 8$$

$$= 25,165,824 \text{ megabits.}$$

Step 3: Calculate transmission time

$$\text{Time} = \frac{\text{Total data in megabits}}{\text{Throughput}}$$

$$\text{Time} = \frac{25,165,824}{120} = 209,715.2 \text{ seconds.}$$

Note:

If the question uses the common simplified convention $1\text{GB} = 1000\text{MB}$, then;

$$3\text{GB} = 3000\text{MB}$$

$$\text{Requests} = \frac{3000}{3} = 1000 \text{ requests}$$

$$\text{Transmission time} = \frac{3 \times 8 \times 1000}{120} = 200 \text{ seconds}$$

Using the usual networking decimal convention, the FTP application generates 1000 requests, and the estimated transmission time is 200 seconds.

⑧

Total End to End Delay :

Given :

Delay	Time
Application Layer	4ms
Transport Layer	3ms
Network Layer	5ms
Data Link Layer	2ms
Physical Layer	1ms
Propagation Delay	18ms
Transmission Delay	12ms

Step 1: Calculate total processing delay

$$\text{Processing delay} = 4 + 3 + 5 + 2 + 1 = 15 \text{ ms}$$

Step 2: Calculate total end-to-end delay

$$\text{Total delay} = \text{Processing delay} + \text{Propagation delay} + \text{Transmission delay.}$$

$$\text{Total delay} = 15 + 18 + 12 = 45 \text{ ms}$$

Contribution of Each OSI Layer:

- * Application Layer - 4ms : Provides network services the user application.
- * Transport Layer - 3ms : Provides end-to-end communication, segmentation, flow control and reliability.
- * Network Layer - 5ms : Creates frames addressing and routing of packets.
- * Data Link Layer - 2ms : Creates frames and provides node to node delivery and error detection.
- * Physical Layer - 1ms : Transmits bits as electrical, optical, or radio signals.
- * Transmission Delay - 12ms : Time required to place the packets bits into the communication link.
- * Propagation Delay - 18ms : Time required for the signal to travel from the sender to the receiver.

9) Protocol Overhead & Transmission Efficiency:

Given:

* Useful data = 8 MB

* Frame size = 1600 bytes

* Protocol overhead per frame = 80 bytes

Step 1: Calculate number of frames

$$8 \text{ MB} = 80 \times 1024 \times 1024$$

$$= 83,886,080 \text{ bytes}$$

$$\text{Number of frames} = \frac{83,886,080}{160}$$

$$= 52,428.8$$

Since a partial frame cannot be transmitted

round up:

52,429 frames

Step 2: Calculate total overhead

$$\text{Total overhead} = 52,429 \times 80$$

$$4,194,320 \text{ bytes}$$

Approximately:

4 MB

Step 3 : Calculate transmission efficiency

Total frame size = 1600 bytes

Useful data per frame: 80 bytes

$$1600 - 80 = 1520 \text{ bytes}$$

$$\text{efficiency} = \frac{1520}{1600} \times 100 = 95\%$$

* Total frames required : 52,429

* Total protocol overhead : 4,194,320 bytes $\approx 4 \text{ MB}$

* Transmission efficiency : boxed {95 %}